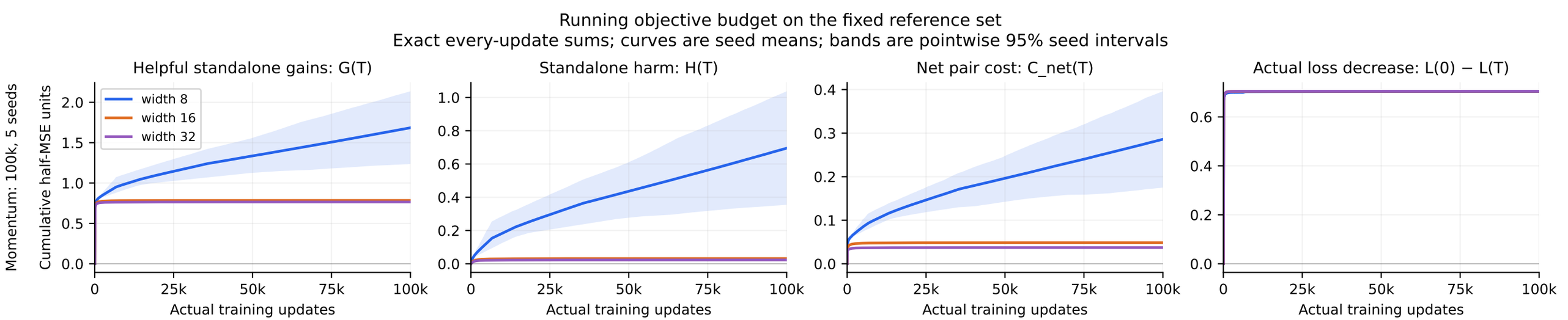

Running objective budget on the fixed reference set
Exact every-update sums; curves are seed means; bands are pointwise 95% seed intervals
Momentum: 100k, 5 seeds
Helpful standalone gains: G(T)
Standalone harm: H(T)
Net pair cost: C_net(T)
Actual loss decrease: L(0) − L(T)
Cumulative half-MSE units
width 8
width 16
width 32
0.0
0.5
1.0
1.5
2.0
0.2
0.4
0.6
0.8
0.1
0.3
0
25k
50k
75k
100k
Actual training updates